

Skills

- **Core ML:** Deep Learning, CNNs, Transformers, Transfer Learning, RAG
- **Programming:** Python, SQL
- **Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn
- **NLP:** spaCy, NLTK, TF-IDF
- **Computer Vision:** OpenCV, YOLO, Pillow
- **MLOps & Tools:** Git, Docker, Linux, Streamlit, LangChain
- **Data:** NumPy, Pandas, Matplotlib, Seaborn

Projects

Moodomic - AI Developer (Major Project)

- Built an AI-based mental health management system integrating a conversational chatbot, CBT modules, and personalized recommendations.
- Implemented a RAG system using LangChain to provide context-aware and personalized responses.
- Designed and managed the knowledge retrieval pipeline to ensure relevant, accurate, and consistent chatbot outputs.
- Worked across the full ML pipeline, including data preprocessing, system integration, and model optimization.

Vecrosoft - AI Developer (Minor Project)

- Developed a CNN-based deep learning model for real-time plant leaf disease detection.
- Designed a custom network architecture and optimized hyperparameters to improve model performance.
- Applied regularization techniques to reduce overfitting, achieving a 3% reduction in error rate.
- Supported early disease detection efforts aimed at minimizing agricultural crop loss.

CO₂ Emission & Temperature Anomaly Analysis ([Link](#))

- Built an interactive dashboard to analyze global CO₂ emissions and temperature anomalies using historical climate datasets.
- Cleaned, processed, and analyzed large-scale data to identify regional and global emission trends.
- Visualized findings through maps, charts to support climate research and policy analysis.
- Recognized as a Global Nominee at NASA Space Apps Challenge 2025 (Dhangadhi) for impactful climate analysis.

Research Papers

PlantNetV1: A Deep Learning-Based Approach for Plant Disease Detection

- Proposed a custom CNN for multi-class crop disease classification
- Presented at Second International Conference on Heritage, Innovation and Transformation (2025)

Suspicious Transaction Detection: DeepDefend, a Multi-Stage Detection Approach

- Designed a layered deep learning pipeline for financial fraud detection
- Presented at First International Conference on Emerging Research Trends and Innovation (2025)

Achievements & Certifications

- **Global Nominee** – [NASA Space Apps Challenge 2025 \(Dhangadhi\)](#)
- **3rd Runner-up** – [Global IME AI/ML Hackathon 2025](#)

- **Python for Everybody Specialization** — University of Michigan ([Coursera](#))
- **Machine Learning Specialization** — DeepLearning.AI ([Coursera](#))
- **Deep Learning Specialization** — DeepLearning.AI ([Coursera](#))

Education

Far Western University

Bachelor of Engineering in Computer Science

Mahendranagar, Nepal

2026(Expected)

Morning Glory Higher Secondary School

Intermediate in Science

Mahendranagar, Nepal

2020